

Elion created the GitReview project

git init

Elion added a **.gitignore** file to ignore unnecessary files or folders

git add .

git commit -m "First commit"

git branch -M main

git remote add origin <https://github.com/NeoTechAcademyQA-7/GitReview.git>

git push -u origin main

Make Sabah a collaborator

Sabah accepted the invitation

Sabah cloned the repository

git clone <https://github.com/NeoTechAcademyQA-7/GitReview.git>

Sabah imported the project to Eclipse

Sabah made some changes to the project, but he didn't commit them

Elion modified a file

Elion added a new file

Elion committed the changes to the local repository

Elion can push the changes by using only -> **git push**

*** Sabah is notified that there is an urgent fix

Stashing is putting your work on hold

Sabah stashed his changes -> **git stash** (**git stash -u** includes untracked files)

Sabah pulled Elion's commit

Sabah finished the urgent fix

Sabah committed & pushed the changes that he made to fix the urgent bug

Now Sabah can continue his work by getting the files and folders from stash

Sabah applied or popped his stashed file -> **git stash pop**

Sabah committed & pushed the changes

To see short form of commits we can use -> **git log --oneline**

git fetch command only fetch the commit from online repository

git pull command applies the commits to my working environment

BRANCHES

Create branches and push the changes

Elion created a branch (online)

git branch -a (shows all the branches, local and online branches)

git fetch (brings the new online branch locally)

git checkout ElionFeature (switch to the branch)

Elion made some changes on the **ElionFeature** branch

Elion pushed the changes to the online **ElionFeature** branch

The purpose of PULL REQUEST is to merge the changes into main branch

Elion created a **Pull Request** (PR) (online)

Elion reviewed and confirmed the Pull Request

The changes are merged to the main branch

*** At this point only the origin main is up to date. And everybody must do a git pull

Elion pulled the **main** branch changes into his local branches

git pull origin main (to update his local **ElionFeature** branch)

Elion switched to **main** branch -> **git checkout main**

git pull (to update his local **main** branch)

Sabah pulled the **main** branch

Sabah created a branch (locally) with this command -> **git branch sabah-branch**

Sabah switched to his local branch -> **git checkout sabah-branch**

Sabah made some changes on his local **sabah-branch**

Sabah pushed the branch online together with the changes

git push -u origin sabah-branch

Sabah created a pull request (PR) (online)

Elion reviewed and confirmed the Pull Request

The changes are merged to the main branch

Elion can get the changes from the online main branch to his local **ElionFeature**

git pull origin main

SCENARIO 1 -> Make changes on different files

Elion made some changes on **ElionFeature** branch
Elion pushed the changes to his online branch
Elion created a Pull Request and merged everything to the main branch

Sabah had local changes on another file
Sabah used **git pull origin main** to update his local branch directly from the origin main
GIT automatically merged the changes (**No need for Sabah to stash changes**)

SCENARIO 2 -> Make changes on the same file

Elion made some changes on **ElionFeature** branch
Elion pushed the changes to his online branch
Elion created a Pull Request and merged everything to the main branch

Sabah had local changes on the same file
Sabah tried to pull the changes from the online main branch -> **git pull origin main**
He got an error (**See the red error below**)
GIT did not automatically merge the changes
Sabah **had to stash his changes**, then he was able to run -> **git pull origin main**
After that Sabah applied the stashed changes -> **git stash pop**

error: Your local changes to the following files would be overwritten by merge:
src/com/neotech/review10/TestCase01.java
Please commit your changes or stash them before you merge.
Aborting

SCENARIO 3 -> Make changes on the same file & on the same lines of code

Resolve the conflict locally

Sabah made some changes on **sabah-branch**

Sabah pushed the changes to his online branch

Sabah created a Pull Request and merged everything to the main branch

Elion made local changes on the same file

Elion tried to pull the changes from the online main branch -> **git pull origin main**

He got an error (**See the red error above**)

GIT did not automatically merge the changes

Elion **had to stash his changes**, then he was able to run -> **git pull origin main**

After that Elion applied the stashed changes -> **git stash pop**

At this moment Elion had a conflict

After resolving the conflict on Eclipse he pushed the changes

Then he merged into online main branch with a Pull Request

SCENARIO 4 -> Make changes on the same file & on the same lines of code

Resolve the conflict online

Sabah made some changes and pushed them to his online branch

Sabah merged the changes to the main branch with a Pull Request

Elion made some changes on the same file and same lines of code

Elion pushed his changes to his online branch

When Elion tried to merge the changes to main branch with a PR, but he got a conflict

After resolving the conflict on GitHub the changes were merged

Deleting the branch

After we are done with our task/feature, we can delete the remote branch

We can also delete our local branch using -> **git branch -D ElionFeature**